



Hardware Components — Input & Output Devices

Grade 10 · ICT · Module 2.2

LEARNING OBJECTIVES

01

Understand

Functions of a computer system

02

Identify

Various input and output devices

03

Distinguish

Soft copy vs. hard copy

04

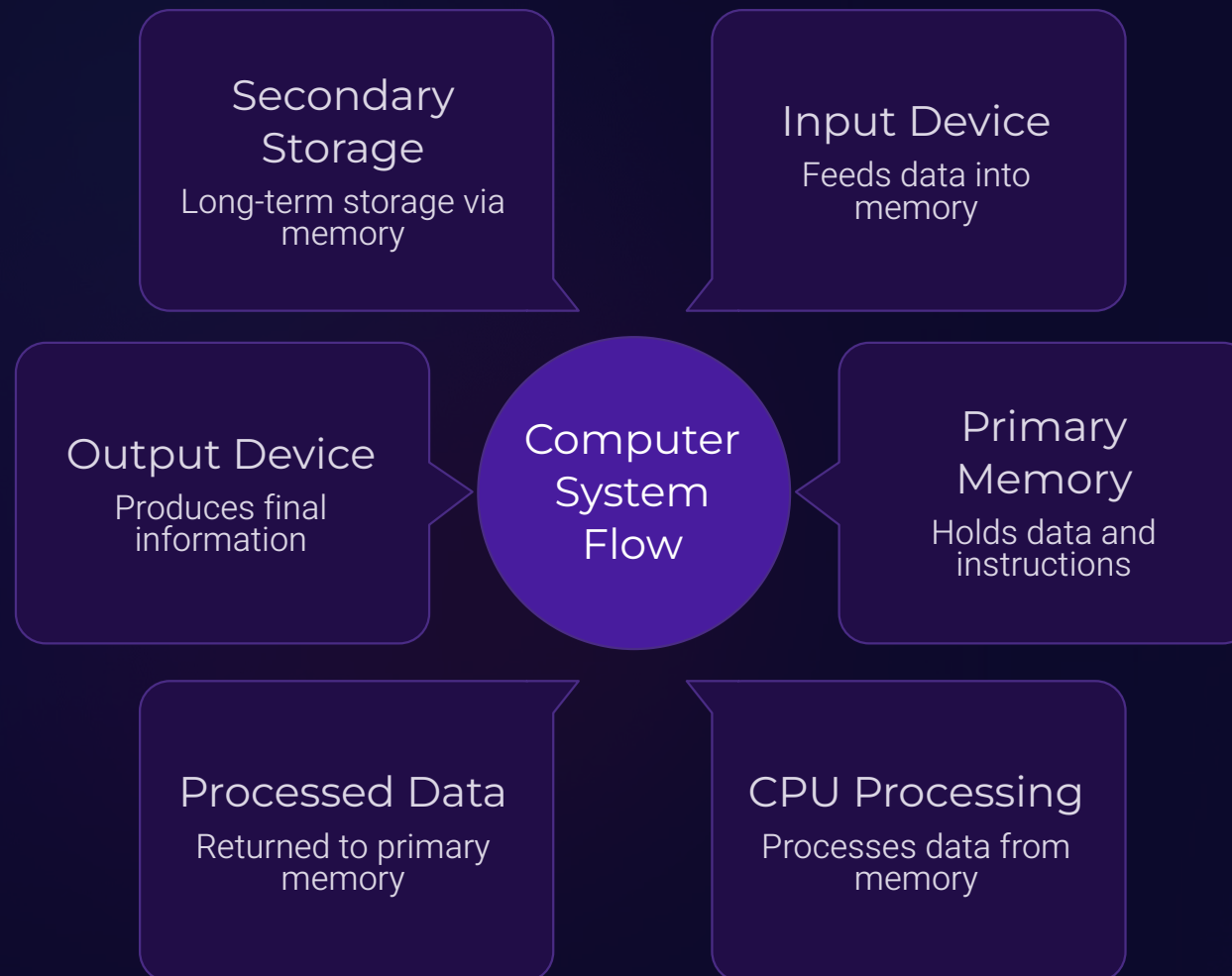
Differentiate

Impact vs. non-impact printers

ICT Grade 10 - Chapter 2 - Module 2 - Akila Savinda

Functions of a Computer System

A computer system performs four main functions: **input data**, **process and store data**, and **produce information** when needed.



Data and instructions are fed via an input device and stored in primary memory.

The CPU processes data using stored instructions.

Processed data becomes information in primary memory, then is sent to the output device.

Data for long-term storage is sent to secondary storage.



Key Point: The Control Unit sends control signals to all devices attached to the computer system.

Input Devices — Introduction

Data and instructions are fed to the computer system using input devices. There are many input devices used for different purposes.



Keyboard

The most common input device for entering text and commands.



Pointing Devices

Mouse, touchpad, touchscreen, joystick, and light pen.



Imaging & Video

Digital cameras, webcams, and CCTV cameras.



Scanners

Flatbed, barcode, MICR, OCR, and OMR scanners.



Other

ATM machines and microphones.

Input Devices — Keyboard



The keyboard is the **most common input device**. Though designed like a conventional typewriter, it provides keys for additional functions. Available in **101/102** and **104/108** key layouts.

Function Keys (F1–F12)

Special commands

Alphanumeric Keys

Letters and numbers

Navigation Keys

Arrows, Home, End,
Page Up, Page Down

Control Keys

Ctrl, Alt, Shift, Esc

Numeric Keypad

Numbers and math operations

Special Keys

Windows, Print Screen, Scroll Lock, Pause Break

Input Devices — Pointing Devices

Pointing devices control the cursor or pointer on the screen.



Mouse

The most popular pointing device. Features left button, right button, and scroll wheel. Available in wired and wireless types.



Touchpad / Trackpad

Built into laptops as a touch-sensitive surface for cursor control.



Touchscreen

Acts as both input and output device. Data is entered by touching the screen. Common in modern computers and mobile phones.



Joystick

Used in computer games, flight simulators, and industrial control systems to move the cursor.



Light Pen

Used to select menu items or draw/write on the screen. Commonly used in Computer Aided Designs (CAD).

Input Devices — Imaging, Video & Scanners



Digital Camera

Inputs photographs and videos to the computer. The display screen serves as the camera's output device.



Webcam

Used for video conferencing, online classes, and streaming. Positioned on top of monitors or connected externally.



CCTV

Captures images and videos for security surveillance and traffic monitoring.



Flatbed Scanner

Functions like a photocopier. Laser beams capture images and convert them to digital data.



Barcode Reader

Reads barcodes and converts them to alphanumeric values. Used in retail, inventory, and libraries.

Input Devices — MICR, OCR, OMR, ATM & Microphone

MICR

Magnetic Ink Character Reader —

Used in banks to read cheque numbers (bank code, account number) printed in magnetic ink.

OCR

Optical Character Recognition —

Captures printed text and converts it to editable digital text. Used in libraries and government institutes.

OMR

Optical Mark Recognition —

Recognizes pencil or pen marks on forms. Commonly used to grade MCQ exam papers.

ATM

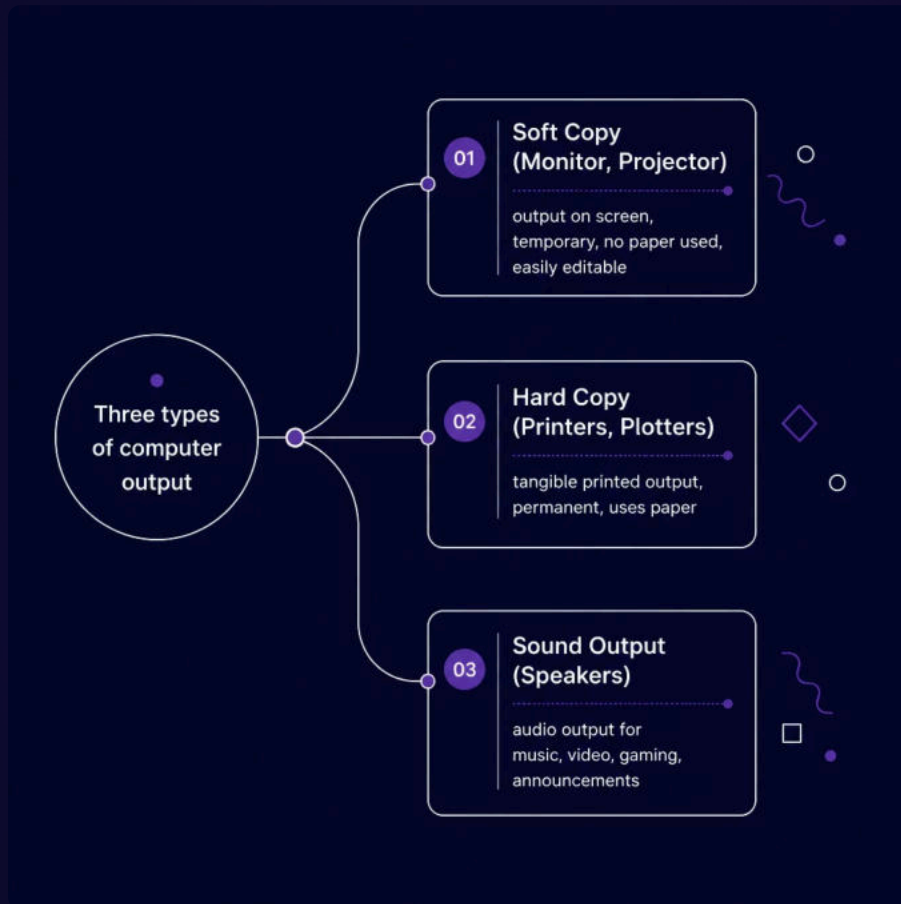
Automated Teller Machine — Reads debit/credit card numbers and connects to the bank's computer for transactions.

Microphone

Feeds sound to a computer. Used for voice recording, voice commands, online communication, and gaming.

Output Devices — Introduction & Soft Copy

There are three types of output: **soft copy** (on screen), **hard copy** (printed), and **sound** (audio).



Soft Copy — Monitor Types

Output on the screen is called a soft copy. The monitor is also known as a Visual Display Unit (VDU).

Projector

Outputs information to a wide screen for multiple viewers. Used in classrooms, meetings, and presentations.

CRT

Large, bulky, high power consumption. Older technology.

LCD

Flat screen, uses liquid crystal. Consumes less electricity.

LED

Best brightness, lowest power consumption. Most modern.

Output Devices — Hard Copy (Printers)

Hard copies are tangible, printed outputs. Printers are classified as **impact** or **non-impact**.

Impact Printers

A head or needle strikes an ink ribbon to mark paper. **Noisy** but inexpensive.

- **Dot Matrix:** Prints as small dots; durable for receipts
- **Line Printer:** Prints one row at a time; ~3,000 lines/min

Non-Impact Printers

Does not strike a ribbon. **Quiet**, better quality, faster.

- **Laser:** Uses laser beam and toner; clear quality print
- **Inkjet:** Liquid ink through nozzles; affordable
- **Thermal:** Uses heat; common for receipts and ATMs

Feature	Impact	Non-Impact
Noise	Noisy	Quiet
Quality	Lower	Higher
Speed	Slower	Faster
Running Cost	Lower	Higher
Examples	Dot Matrix, Line	Laser, Inkjet, Thermal

Module 2.2 Summary

Input Devices

- **Keyboard** — Text and commands
- **Pointing** — Mouse, touchpad, touchscreen, joystick, light pen
- **Imaging/Video** — Camera, webcam, CCTV
- **Scanners** — Flatbed, barcode, MICR, OCR, OMR
- **Other** — ATM, microphone

Output Devices

- **Soft Copy** — CRT, LCD, LED, projector
- **Impact Printers** — Dot matrix, line printer
- **Non-Impact** — Laser, inkjet, thermal
- **Plotter** — Architectural and design drawings
- **Sound** — Speakers

 **Activity:** Observe and identify the input and output devices in your computer laboratory.